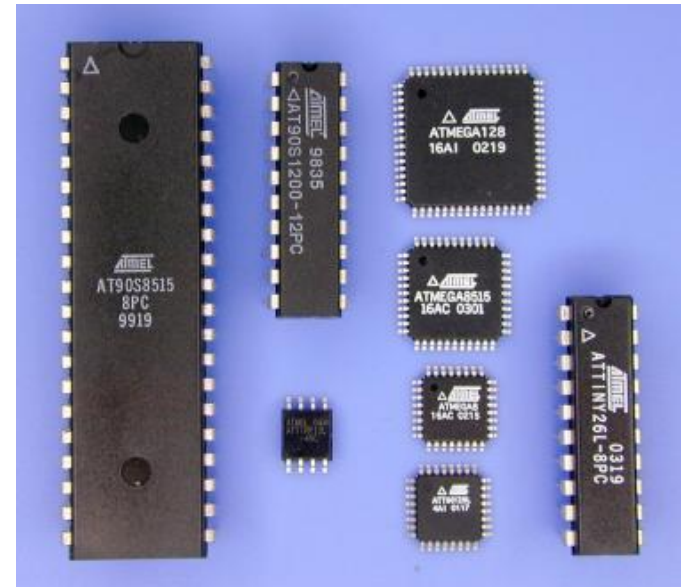


Microprocessors and Microcontrollers



What is a Microcontroller?

- What is a microcontroller?
 - microprocessor
 - Read-write memory
 - Flash memory
 - EEPROM
 - Input/Output Control
 - Peripheral Devices (AD/DA)



What is a Microcontroller?

- Microcontroller is just like a Computer we use everyday, but only in a smaller scale
- Contains most of the basic features of a Computer System in a single IC
- Can take input
- Can provide output
- Can perform mathematical calculations and logical operations
- And can do much more things



Microcontroller and PC

	PC	uC
CPU Clock	3 GHz	20 MHz
RAM	512 MB	1 kB
Hard disk	80 GB	16kB
Processor	32/ 64 bit	8 bit
OS	Windows	Not required
Input	KB, Mouse	Keypad
Output	Monitor, printer	LCD module

Why use a uC

- Emphasis is on cost
- Emphasis on power consumption
- Easy to use- small package
- Enough on-chip-resources
- Efficient for high level languages
- Dedicated application

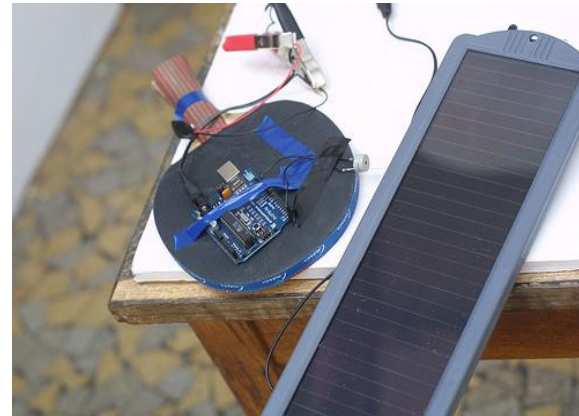


Uses of uC

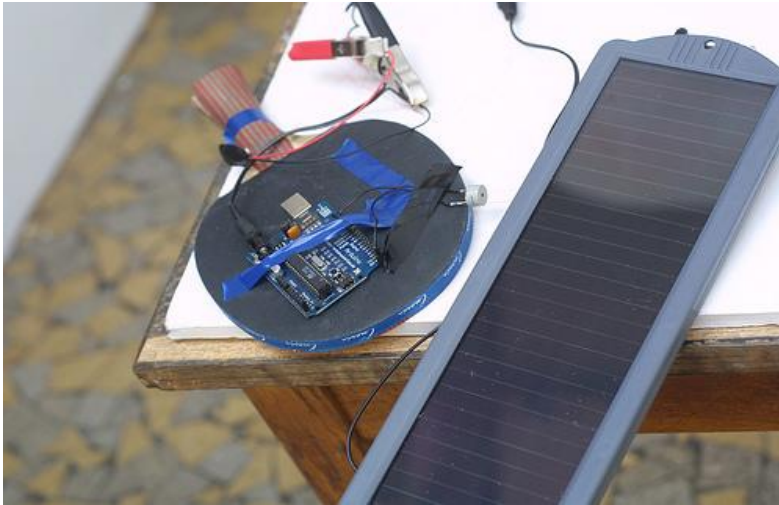
- Control systems & Appliances
- Robotics
- Consumer electronics
- Measuring devices
- Lock Devices
- Hobbyist projects!
 - Musical instruments
 - Home automation



Uses of uC



Other Examples



Different Families of Microcontroller

- 8051
- AVR
- PIC



